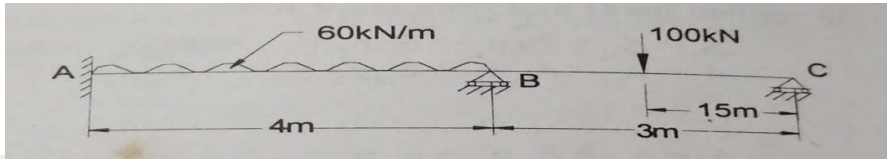
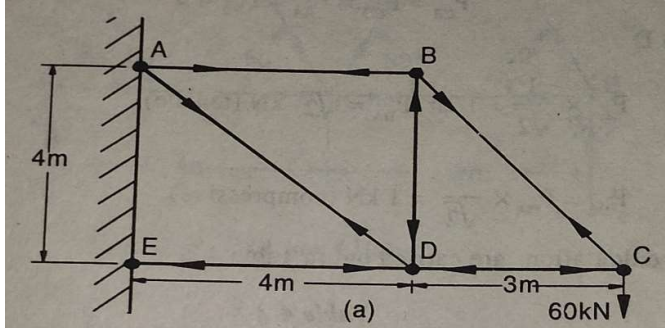
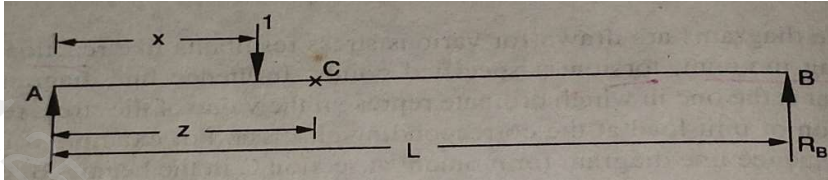


DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2024 Course: B. Tech. Branch : Civil Engineering Semester : V Subject Code & Name: BTCVC503 Structural Mechanics-II Max Marks: 60 Date: 05/07/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/ CO)	Marks
Q. 1	Solve Any Two of the following		
A)	State and explain Muller-Breslau principle	CO2	06
B)	Compare flexibility and stiffness methods.	CO4/5	06
C)	State and explain application of Eddy's theorem.	CO3	06
D)	Write a short note on 'Finite element method'	CO5	06
Q.2	Solve Any ONE of the following.		
A)	Compute the ordinates of influence lines for reaction R_a for the beam as shown in figure.1 at 1m interval and draw the influence line diagram. The M.I is constant uniform throughout.	CO2	12
	Fig.1		

B)	Analyze the beam as shown in figure2, by Flexibility matrix method	CO5	12
 Fig.2			
Q. 3	Solve Any ONE of the following.		
A)	Determine the horizontal thrust developed in a two-hinged semi-circular arch subjected to a u.d.l on only one-half of the arch. EI is constant throughout.	CO4	12
B)	A three-hinged stiffening girder of a suspension bridge of span 100 m is subjected to two point loads of 200 kn and 300kn at a distance of 25 m and 50 m from the left end. find the SF and BM for the girder at a distance 30 m from the left end. The supporting cable has a central dip of 10 m.find also maximum tension and its slope in the cable.	CO3	12
Q.4	Solve Any ONE of the following.		
A)	Determine the vertical deflection of point D in the truss as shown in figure 3. Cross sectional area for members AD & DE are 1500 mm ² and for remaining members is 1000 mm ² . Take E=200GPa.	CO1	12
 Fig.3			
B)	Analyze the beam as shown in figure 2, by stiffness method. Take EI constant throughout	CO4	12
Q. 5	Solve Any One of the following.		

A)	Draw ILD for the reaction R_a, SF at C moment at C for the beam as shown in figure 4.  Fig.4	CO6	12
B)	A three-hinged parabolic arch with supports at same level, 24 m in span with 4 m central rise is subjected to a u.d.l of 30 kN/m on left half portion and a point load of 50 kN at a distance of 18m from left support throughout its span . Calculate the moment, thrust and radial shear at a section 6m from the left support.	CO2	12
	*** End ***		